MIL-P-63105B(AR)
24 June 1980
SUPERSEDING
MIL-P-63105A(AR)
15 November 1978

MILITARY SPECIFICATION

PROPELLANT M30 FOR USE IN CARTRIDGE 105MM, APFSDS-T, M735, M735A1, AND XM774

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M30 Propellant for use in the cartridge APFSDS-T, M735, M735A1, and XM774 for the 105MM, M68 Gun.

1.2 Classification.- The propellant shall be Type I, cylindrical multi-perforated grains.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 -Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-286	-Sampling Procedures and Tables for Inspection by Variables for
MIL-STD-652	-Propellants, Solid, for Cannons, Requirements and Packing
MIL-STD-414	-Sampling Procedures and Tables for Inspection by Variables for percent defective

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9296617	-Primer, Electric, M120, Loading Assembly
9312500	-Cartridge, 105MM, APFSDS-T, M735
9326059	-Cartridge, 105MM, APFSDS-T, M735A1
9329513	-Cartridge, 105MM, APFSDS-T, XM774
10522799	-Case Cartridge, 105MM, M148A1B1

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material.- The propellant shall be M30, conforming to the requirement of MIL-STD-652 and Cartridge drawing applicable (see 2.1) except for the following:

3.1.1 Grain length to diameter ratio.- The average grain length (L) shall be from 2.00 to 2.20 times the average grain diameter (D).

3.1.2 Grain diameter to perforation diameter ratio.- The average grain diameter (D) shall be from 11.5 to 15 times the average diameter of the perforation (d).

3.1.3 Web.- The nominal web size shall be 0.051 inch (in.).

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3.1.4 Bulk density.- The nominal bulk density shall be 63.0 pounds per cubic foot (lb/ft^3). (See 4.5.2)

3.1.5 Graphite content.- The graphite (Glaze) content shall not exceed 0.2 percent. (See 4.5.3)

3.2 Ballistic assessment requirements.- The propellant when fired in the M68 Gun with the projectile M735, M735A1, or XM774 (see 4.5.4) shall comply with the following requirements at 70 degrees Fahrenheit (^oF) and standard weapon conditions. The propellant shall be capable of assessment to a velocity of 4925 feet per second (fps) with the M735 and M735A1 Projectile, and 4950 fps with the XM774 Projectile. The recommended charge weight shall not exceed 224 ounces. (See 4.5.1.1)

3.3 Ballistic uniformity requirements. (IMP) The propellant when fired in the M68 Gun with the projectile M735, M735A1, or XM774 shall comply with the following requirements. (See 4.5.1.2)

3.3.1 Uniformity of velocity.- The standard deviation of velocity for the lot at 70^oF shall not exceed 9.6 fps.

3.3.2 Average pressure.- At 70^oF and standard weapon conditions, the propellant shall not produce an average pressure greater than 60,500 pounds per square inch (psi).

3.3.3 Individual maximum pressure. (IMP)- At 145^oF and standard weapon conditions.

3.3.3.1 The propellant shall not produce an individual maximum pressure greater than 75,700 psi.

3.3.3.2 The average pressure produced by the propellant plus three standard deviations shall not exceed 75,700 psi.

3.4 Preproduction inspection.- This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples by the contractor shall be as specified by the Contracting Officer.

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3.5 Workmanship.- The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.- Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.- The following types of inspection shall be conducted on this unit of product:

a. Preproduction Sample Inspection
b. Quality Conformance Inspection

4.3 Preproduction sample (pilot lots).- The preproduction sample is intended to provide the necessary information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1 Submission.- The Contractor shall select three propellant granulations, which, from experience, are expected to bracket the requirements of this specification. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2, consisting of three pilot lots of a maximum of 3000 lb each, representing the three candidate granulations.

4.3.2 Inspections to be performed.- Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

4.3.3 Rejection.- Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select the one pilot lot which best fits the requirements of this specification. The Technical

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Agency shall notify the Contracting Officer of the selected granulation. The Contracting Officer shall designate the chosen pilot lot as the accepted preproduction sample and shall authorize the Contractor to produce the selected granulation. If none of the pilot lots submitted meets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. Until the preproduction sample is accepted, the Contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the Contracting Officer. All future production lots should be of the same basic granulation as the accepted pilot lot.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.- Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and testing.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2.1 Sampling for laboratory testing (see 3.1).- Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652.

4.4.2.2 Sampling for ballistic testing (see 3.2).- Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652.

4.4.2.3 Charge establishment (see 3.2).- If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify DRDAR-QAR-R and DRSAR-QAM.

4.4.2.4 Velocity uniformity (70°) (see 3.3.1).- If the product of the velocity standard deviation times the factor 0.74 of the rounds tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply retest criteria in accordance with 4.5.1.3. If the retest criteria cited do not apply, the lot shall be rejected.

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4.4.2.5 Average pressure (see 3.3.2).- If the average pressure obtained exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.4. If the referral criteria cited do not apply, the lot shall be rejected.

4.4.2.6 Individual maximum pressure.

4.4.2.6.1 If any individual pressure observed during the uniformity series, when corrected to standard conditions and recommended charge weight exceeds the applicable requirement, the lot shall be rejected.

4.4.2.6.2 If the average pressure plus three standard deviations exceeds the applicable requirement, apply retest criteria in accordance with 4.5.1.3. If the retest criteria cited do not apply, the lot shall be rejected.

4.5 Test methods and procedures.

4.5.1 Proving ground assessment. - These tests shall be conducted at a Government proving ground (designated by the Contracting Officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. Tubes used for these assessments shall be restricted to firing only Armor Piercing Fin Stabilizing Discarding Sabot with Tracer (APFSDS-T) ammunition. The wear of the tube shall be no greater than 0.015 in. when measured 25.25 in. from the rear face of the tube. All test rounds shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made, when fired, that the charges are at the conditioning temperature. The following tolerances shall apply to all temperature conditionings: $-50^{o}F + 5^{o}F$; $70^{o}F \pm 2.5^{o}F$; $145^{o}F - 5^{o}F$. Copper crusher gages (2) shall be used to measure the maximum chamber pressure. Skyscreens will be used to measure the muzzle velocity. Test rounds shall be assembled in accordance with the applicable drawing.

4.5.1.1 Charge establishment (see Table 1).

4.5.1.1.1 Charge slope method.- Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at $70^{o}F$. Three calibration rounds shall also be conditioned and fired at

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70°F, prior to the test rounds. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.1.2.1). The difference between the corrected velocity during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. The resultant will be the recommended charge weight.

4.5.1.1.2 Constant slope method.- After ten lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight shall be determined by the constant slope method. A composite charge weight versus velocity curve can be established from test results generated. This curve can then be used as in 4.5.1.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for th uniformity series and final adjustment is to be made as indicated above for the charge slope method (see 4.5.1.2.2). The resultant will be the recommended charge weight.

4.5.1.2 Uniformity (see Table I).

4.5.1.2.1 Initial production.- Thirty rounds shall be assembled with the calculated charge weight (see 4.5.1.1.1) of propellant. Ten rounds each shall be conditioned to -50°F, 70°F and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 4.5.4) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. Testing at -50°F shall be for information only. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification.

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4.5.1.2.2 Subsequent production.- Twenty rounds shall be assembled with the calculated charge weight (see 4.5.1.1.2) of propellant. Ten rounds each shall be conditioned to 70°F and 145°F. Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at both temperatures shall be corrected to standard projectile weight and recommended charge weight conditions (i.e., the differences between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. Throughout production, if 3 consecutive lots fail on first test, testing shall revert back to the initial production plan.

4.5.1.3 Retest criteria.- All original and retest data shall be submitted to US Army Armament Research and Development Command, ATTN: DRDAR-QAR-R, Dover, New Jersey 07801. A lot shall be retested whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. The final adjustment in charge weight exceeds 2.0 oz. of the charge weight used in uniformity firings retest for uniformity in accordance with 4.5.1.2.

b. If for any reason, the Proving Ground considers that test conditions have detrimentally affected test results, additional charges as required shall be tested.

4.5.1.4 Referral criteria.- A test lot shall be referred to the procuring activity for disposition whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds 25 fps, after allowance has been made for ballistic hump. Ballistic hump is characterized by a rise in velocity and pressure during the early life of a barrel.

c. Number of test or calibration results utilized in calculations is less than specified.

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d. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

e. Test lot fails velocity standard deviation or average pressure requirement at 70°F or 145°F and any extreme test value is an outlier, retest for uniformity. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_C) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

X_1 = Test result being tested as low outlier
X_2 = Test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = Test result adjacent to the largest test result
X_{10} = Test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_C.

4.5.2 Bulk density.- The bulk density shall be determined in accordance with Method 502.2.1 of MIL-STD-286 except that a sample size of approximately 0.25 cubic feet shall be used and 1000 milliliter graduate cylinder shall be replaced with an M148A1B1 cartridge case Dwg. 10522799 with M120 Primer Dwg. 9296617.

4.5.3 Graphite (Glaze).- The graphite content shall be determined in accordance with Method 308.1 of MIL-STD-286.

4.5.4 Projectile weight.- The average projectile weight for the M735 Projectile shall be 12.78 ± 0.02 lb. The average projectile weight for the XM774 Projectile shall be 12.70 ± 0.02 lb (see 6.6).

5. PACKAGING

5.1 Packing and marking.- Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A (see 6.5).

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TABLE 1

PROVING GROUND TEST SUMMARY

Test Phase	Projectile	Temp (°F)	Sample Size		Velocity Requirements (fps)		Pressure Requirements (psi)	
			cal	test	service	std dev	imp	average
Charge Establishment	M735 and M735A1	70	3	9	4925	N/A	N/A	60,500
	XM774	70	3	9	4950	N/A	N/A	60,500
Uniformity	M735 and M735A1	70	10	10	4925	9.6	N/A	60,500
	XM774	70	10	10	4950	9.6	N/A	60,500
	M735, M735A1 and XM774	-50*	N/A	10	N/A	N/A	N/A	N/A
	M735, M735A1 and XM774	145	N/A	10	N/A	N/A	75,700	**

*performed for initial production testing only

**the average pressure plus three standard deviations shall not exceed the imp.

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6. NOTES

6.1 Ordering data.- Ordering data shall be in accordance with MIL-A-48078 with the following exceptions:

6.1.1 Procurement requirements.- Procurement requirements of MIL-A-48078 shall apply.

6.1.2 Contract data requirements.- One copy of each of the Propellant Description Sheet and the Acceptance Test Summary shall be forwarded to each office designated by the Contracting Officer.

6.1.3 Charge weight.- If the recommended charge weight established for a cartridge is to be used in a different cartridge, the charge weight shall be corrected for the difference in projectile weight.

6.2 Proving ground test summary (see Table I).

6.3 Drawings.- Drawing listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.4 Surveillance sample.- Samples shall be forwarded to US Army Armament Research and Development Command, ATTN: DRDAR-LCE-MP, Dover, New Jersey 07801 for the propellant surveillance test program in accordance with MIL-STD-652.

6.5 Reference sample.- Whenever a propellant lot is selected for reference use, a 5-lb sample shall be forwarded to US Army Armament Research and Development Command, ATTN: DRDAR-LCA-G for use in evaluating the propellant stockpile.

6.6 Propellant Assessment.- Propellant Assessment may be required for a specific production lot of projectiles whose average as fired weight has been found to significantly deviate from the standard projectile weight for reasons other than projectile design modification.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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